

EUGENE L. BUCHANAN

Staff AI Engineer — Multi-Agent Frameworks & Agentic Infrastructure

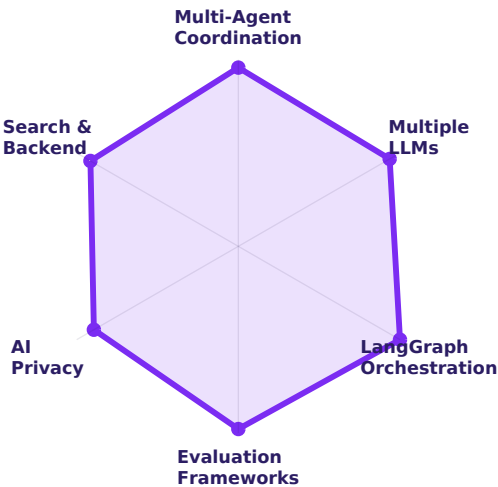
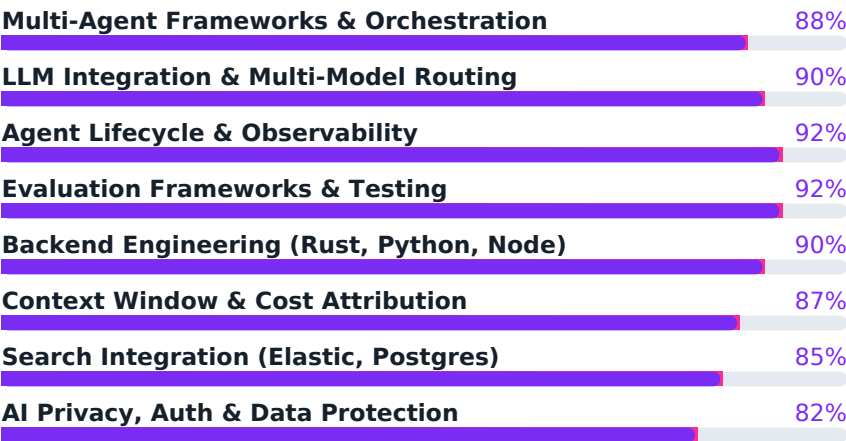


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Professional Summary

Staff AI engineer with 27 years software/systems experience focused on agentic AI infrastructure and multi-agent orchestration. Three production PRs to ZeroClaw (32.2k★ Rust gateway): merged context-window meter for 9 LLM providers, open PR for live provider identity on usage events (cost attribution), agent lifecycle observer via JSON-RPC. Built test automation, integration platforms, CI/CD pipelines on AWS/Postgres/React. Passionate about automatic model routing and platforms for deploying intelligent agents at scale.

Technical Expertise



Technology Stack

- LangGraph Multi-Agent Coordination Agent Lifecycle JSON-RPC Bounded I/O Fire-and-Forget
- Unix Domain Sockets
- Context Window Routing Multi-Model Cost Attribution MCP (Model Context Protocol) herdr-mcp Server
- External Tool Access Prompt Engineering OpenAI Anthropic Cohere Gemini
- Rust Python Node.js PostgreSQL AWS (ECS, S3) REST GraphQL Docker Git
- Cypress Selenium JUnit Postman Elasticsearch Regression Suites Health Dashboards Alerting
- SAML OAuth OIDC Permission Profiles Principal Isolation Deny-by-Default

Experience

Staff AI Engineer / Agentic Infrastructure Contributor (Open-Source)

2026 – Present

[ZeroClaw Labs \(github.com/zeroclaw-labs/zeroclaw\)](https://github.com/zeroclaw-labs/zeroclaw) · Rust · 32.2k stars

- PR #7946 (MERGED) — feat(runtime): model context window ctx bar across TUI/gateway/CLI (+1,109 / -22, 25 files). context_window as single source of truth for 9 providers; auto-populated config, doctor command, gateway API, live ContextBar. Eliminated drift between surfaces — directly relevant to Brain2's automatic model routing.
- PR #8966 (OPEN) — feat(agent): carry live provider identity on usage events (+2,742 / -141, 20 files). Fixed frozen 32k meter on 1M-token models; separated trim budget from model capacity; unconditional Usage events with serving-provider identity; per-provider usage breakdown on done frame — cost-attribution infra for multi-LLM orchestration.
- PR #8337 (OPEN) — feat(observability): herdr agent reporting integration (+1,712 / -17, 13 files). Observer reports agent lifecycle (idle/working/blocked/released) via JSON-RPC over UDS. Zero-config env detection, approval-gate blocked state, crash recovery via pane.release_agent. Multi-agent coordination infrastructure; maps to Super Agents' human-agent collaboration.

Senior Software Engineer / QA Lead / Integrations Product Owner

Dec 2020 – Aug 2025

[Knowledgecity LLC](#) · Remote · AWS / Postgres / React

- Product Owner, Integrations: shipped SAP, Oracle, Workday, Coursera, UKG, and Zoom integrations via SAML, OAuth, SFTP, and custom REST APIs — orchestrating multiple services with different auth models, data formats, and error semantics, the same coordination challenge as multi-agent systems.
- Built and maintained CI/CD quality pipelines with Cypress, Selenium, Postman, JUnit, Elastic, S3 — an evaluation framework for testing complex deployment scenarios across integration, backend, API, database, and frontend surfaces.
- Applied AI prompt engineering and chatbot support in production, integrating LLM-powered tooling into customer-facing assistance and internal automation workflows.
- Designed and implemented evaluation frameworks for regression suites, health dashboards, and alerting that measured system-level dynamics across multi-environment deployments.
- Mentored and trained test automation team; conducted code reviews and elevated engineering standards.

Senior Software Engineer II / Presales Engineering

1999 – 2001

[RealNetworks](#) · Seattle, WA · Streaming Media Platform

- QA on advanced research team focused on consumer appliances, mobile platforms, stream servers, and cellular networks — evaluating distributed system behavior in complex, multi-component scenarios.
- Global presales engineering: translating product capabilities into customer solutions across diverse technical environments. New-hire training and cross-functional collaboration.

Senior Software Test Engineer IV

1997 – 1999

[Microsoft](#) · Redmond, WA · Windows Media Server

- White-box testing/automation for Windows Media Server, Windows 98/NT — evaluation frameworks for complex system testing.
- Deep debugging across server infrastructure, network protocols, media codecs.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International

Key Strengths

- **Multi-agent orchestration.** Shipped agent lifecycle observer (PR #8337) + MCP server in 32k-star Rust gateway — not a demo.
- **Multi-model routing & cost.** Merged context-window meter for 9 LLMs (PR #7946); open PR for live provider identity on usage events (PR #8966).
- **Evaluation frameworks.** 27 years test automation, regression suites, quality pipelines for complex distributed systems.
- **LLM integration.** Prompt engineering + chatbot tooling in production; multi-provider orchestration daily.
- **Cross-functional collaboration.** Product Owner, QA Lead, presales engineer — comfortable with PMs, designers, engineers.